When Data Quality Defects Interact: Measuring the Impact of Missing Values and Duplicate Leakage on Machine Learning Evaluation

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***Abstract*—*Reliable supervised machine learning evaluation relies on the fundamental assumption that training and test partitions are independent and identically distributed (i.i.d.). However, real-world data pipelines frequently suffer from co-occurring data quality defects. In particular, duplicate records crossing partition boundaries introduce test-to-training leakage, while feature missingness alters the information content of those observations. Although existing studies treat data leakage and missing values as separate operational challenges, their joint empirical interaction remains poorly understood. This study systematically investigates this interaction across three tabular benchmark datasets: Wine Quality—Red (N = 1,599), Wine Quality—White (N = 4,898), and a stratified 10,000-record subsample of Adult Census Income (N = 10,000). We evaluate three foundational classifiers—Logistic Regression, Decision Trees, and Random Forests—across four Missing Completely at Random (MCAR) feature-masking levels (0%, 10%, 20%, 30%) and four controlled test-to-training duplicate overlap levels (0%, 10%, 25%, 50%), alongside ordinary random splitting, across ten random experimental seeds (2,160 model evaluations). Our findings demonstrate that duplicate overlap produces severe, model-dependent performance inflation: at 50% overlap and 0% additional missingness, F1-scores were inflated by +0.142 to +0.202 for Decision Trees and +0.098 to +0.168 for Random Forests ($p < 0.002$, Wilcoxon signed-rank tests), eliminating 52.2% to 60.8% of apparent reference classification errors (Error-Reduction Illusion). In contrast, Logistic Regression exhibited negligible sensitivity ($\Delta F1 \leq +0.006$). Under ordinary random splitting, natural duplicate prevalence (15.0% in Red Wine and 19.1% in White Wine) caused statistically significant F1 inflation of +0.035 to +0.066 in tree models ($p < 0.05$), while producing no inflation in Adult (0.05% duplicates). Furthermore, ordinary least squares (OLS) interaction models reveal that missingness significantly amplifies tree-based duplicate leakage in higher-dimensional data ($\beta_inter = +0.244$, $p < 10^{-9}$ for Random Forest on Adult). Finally, a supplementary mechanism experiment shows that independently masking duplicate copies destroys the exact coordinate alignment required by trees, reducing the retained leakage effect to only 15.0%–32.3% at 30% missingness. We formulate rigorous, pipeline-level recommendations to audit duplicate structures and enforce leakage-safe partitioning in scientific and industrial workflows.*

***Keywords*—**Data quality, missing values, duplicate leakage, machine learning evaluation, data partitioning, error-reduction illusion.

I. INTRODUCTION

Supervised machine learning algorithms are increasingly deployed in mission-critical decision systems, including clinical risk assessment, automated credit underwriting, fraud detection, and predictive maintenance. The scientific and operational credibility of these models rests upon empirical evaluation: fitting candidate models on a designated training subset and assessing their generalisation performance on an independent, held-out test partition. The mathematical validity of this paradigm assumes that training and test samples are independently and identically distributed (i.i.d.) realizations drawn from a stationary data-generating distribution [1].

In practice, this independence assumption is routinely violated by data leakage—the unintended introduction of information from the test set (or downstream evaluation context) into the training pipeline [1], [5]. A prevalent yet insidious vector of data leakage is duplicate records. Repeated measurements, sensor logging anomalies, multi-source record aggregation, and customer retrials frequently inject identical or near-identical observations into tabular datasets. When a naive random split is applied to a dataset containing duplicate instances, identical feature-label tuples cross the train-test boundary. Consequently, an evaluation metric computed on the test partition ceases to measure generalisation to unseen observations; instead, it measures the capacity of the model to memorize and retrieve previously encountered points [1], [4].

Simultaneously, real-world tabular data suffer from another ubiquitous quality defect: missing values. Observational non-response, sensor failure, communication dropouts, and multi-modal integration gaps result in incomplete feature vectors [2], [3]. While a substantial body of literature has benchmarked imputation techniques across diverse missingness mechanisms [3], missing-data research has predominantly operated under the assumption of leak-free data partitioning. Conversely, data leakage studies have largely operated on complete, uncorrupted feature spaces [1].

In practical machine learning lifecycles, these two defects do not occur in isolated silos; they interact. Specifically: Does feature corruption attenuate the memorization advantage gained from duplicate leakage? Or conversely, does imputation provide an artificial anchor that preserves or even exacerbates the distortion of evaluation metrics? How does classifier inductive bias—ranging from linear hyperplanes to non-parametric recursive partitioning—modulate this vulnerability? And what is the empirical consequence of standard random train-test splitting when tabular datasets possess naturally occurring duplicates?

To address these foundational questions, this study conducts an exhaustive, multi-dataset, multi-model empirical investigation into the interaction between missing-value

contamination and duplicate leakage. The main contributions are: (1) Quantifying the quantitative sensitivity of linear baselines and flexible tree models across controlled duplicate overlap levels (0% to 50%), establishing the Leakage Sensitivity Index (LSI) and Error-Reduction Illusion (ERI); (2) Assessing whether feature missingness moderates or amplifies duplicate leakage via OLS interaction models; (3) Auditing natural duplicate leakage in standard random splitting across contrasting datasets; (4) Disentangling feature identity from instance duplication via an independent-masking experiment; and (5) Delivering actionable, pipeline-level safeguards for robust machine learning workflows.

II. RELATED WORK AND RESEARCH GAP

A. Data Quality in Machine Learning

Data quality is a multi-dimensional construct encompassing accuracy, completeness, consistency, timeliness, and uniqueness [4]. Mohammed et al. [4] investigated how data-quality dimensions (including missingness, label noise, and feature noise) degrade the performance of supervised learning algorithms on tabular benchmarks. Their findings indicated that algorithmic resilience varies markedly across architectures, with tree ensembles often exhibiting greater robustness to noise than linear or distance-based algorithms [6], [8]. However, their benchmarking evaluated models under conventional partitioning schemes, without explicitly modeling the confounding effects of cross-partition duplicate leakage.

B. Missing-Value Handling

Missing data theory categorizes missingness into Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR) [16]. Emmanuel et al. [2] reviewed missing-data handling techniques, spanning listwise deletion, mean/median statistical imputation, k-nearest neighbors (KNN), and generative modeling [10]. Jäger et al. [3] benchmarked 14 imputation algorithms across multiple tabular datasets and missingness proportions, demonstrating that simple median/mode imputation remains a highly competitive and computationally efficient baseline, particularly when downstream models are decision-tree ensembles [6], [9]. Crucially, existing missingness benchmarks fit imputers strictly within clean train-test splits, leaving unaddressed how missingness behaves when identical observations reside on both sides of the evaluation split.

C. Data Leakage and Partitioning Strategies

The reproducibility crisis in machine learning has highlighted data leakage as one of the primary drivers of irreproducible claims [1]. Kapoor and Narayanan [1] surveyed over 300 papers across 17 fields, discovering widespread methodological leakage that led to severe performance overestimation. They categorized leakage into pre-split preprocessing leakage, feature leakage, and duplicate/distributional leakage [5], [19], [20]. Kaufman et al. [5] and scikit-learn guidelines [15] emphasized encapsulating preprocessing within training-only pipelines. However, the interaction between duplicate contamination across partitions and feature corruption within pipelines remains unexplored.

D. Research Gap and Research Questions

Although the literature addresses data quality, missing data, and leakage in isolation, their joint interaction represents an unmapped crossroad in empirical machine learning methodology. Specifically, does missingness obscure duplicate records and eliminate leakage, or does it amplify the vulnerability of evaluation metrics by degrading the true signal? This investigation resolves four formal research questions:

- RQ1: How does controlled duplicate overlap affect the F1-score across classifiers possessing distinct inductive biases (linear vs. single-tree vs. ensemble)?
- RQ2: Does increasing feature missingness systematically moderate (dampen or amplify) the magnitude of duplicate-leakage effects?
- RQ3: To what extent does ordinary random splitting inflate performance estimates on tabular datasets with varying degrees of natural duplicate prevalence?
- RQ4: Does duplicate leakage stem from the presence of repeated target-correlated instances, or does it strictly require the preservation of exact, multi-dimensional feature coordinates?

III. METHODOLOGY

A. Research Design

We executed a controlled, full-factorial experimental design implemented in Python using scikit-learn [15], numpy, pandas, scipy, and statsmodels. The design encompasses: 3 benchmark datasets (Wine Quality—Red, Wine Quality—White, Adult Census Income subsample); 3 classification models (Logistic Regression, Decision Tree, Random Forest); 4 additional MCAR feature cell masking levels (0%, 10%, 20%, 30%); 4 controlled overlap conditions (0%, 10%, 25%, 50%); 1 natural ordinary random split condition; 1 supplementary independent-masking condition at 50% overlap; and 10 random seeds (0 to 9). In total, the experimental suite executes 2,160 fully trained and evaluated classification pipelines.

B. Datasets and Duplicate Auditing

We selected three open-access tabular benchmarks from the UCI Machine Learning Repository [11], [12] to reflect

contrasting dimensionalities, feature typologies, and intrinsic duplicate structures (Table I).

TABLE I. DATASET CHARACTERISTICS AND NATURAL DUPLICATE AUDIT

Dataset	Records	Features	Duplicates	Target Class	Pos. Ratio
Wine Red [11]	**1,599**	**11 num**	**240 (15.0%)**	**Quality ≥ 6**	**53.5%**
Wine White [11]	**4,898**	**11 num**	**937 (19.1%)**	**Quality ≥ 6**	**66.5%**
Adult Income [12]	**10,000**	**6 num, 8 cat**	**5 (0.05%)**	**Income > 50K**	**23.9%**

Wine Quality datasets comprise physicochemical measurements of Portuguese Vinho Verde wines [11]. The positive class is defined as sensory quality score ≥ 6. Adult Census Income is extracted from the 1994 US Census [12]; we used a 10,000-record stratified subsample (seed 26) with pre-existing missing entries preserved. Automated auditing verified zero contradictory duplicate rows (identical features with conflicting labels) across all datasets.

C. Missing-Value Injection and Pipeline Encapsulation

Feature missingness is injected under an MCAR mechanism at rates m ∈ {0.0, 0.10, 0.20, 0.30}. Target labels are never masked. To strictly prevent preprocessing leakage [5], all transformations are encapsulated in a scikit-learn Pipeline. Numerical columns undergo training-set median imputation followed by StandardScaler. Categorical columns undergo training-set mode imputation followed by OneHotEncoder. All parameters are fitted strictly on the training partition.

D. Reference and Leakage Partitioning Schemes

The reference split employs a stratified group-aware partitioner. Every unique record (feature tuple plus label) is assigned an indivisible group ID, ensuring that training and testing partitions share zero identical observations (0% overlap). Controlled overlap conditions (10%, 25%, 50%) select K = round(α · |X_test|) observations from the test set and append exact copies to the training partition, evaluating on the original test set. The ordinary random split condition applies stratified row-level splitting without grouping.

E. Supplementary Mechanism Experiment

Fixed at 50% overlap, the supplementary experiment compares exact duplicate copies against independently masked copies. In the exact condition, training copies share the identical mask pattern as their test counterparts. In the independent condition, training copies are drawn from unmasked records and corrupted independently with rate m. This isolates whether tree models benefit from general neighborhood density or strictly from exact coordinate matching.

F. Classification Models and Evaluation Metrics

We evaluate Logistic Regression (max_iter=2000), CART Decision Tree (unpruned), and Random Forest (100 trees, sqrt features). Primary metric is positive-class binary F1-score:

$$F1 = (2 \cdot \text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (1)$$

We derive F1 inflation (ΔF1 = F1_contaminated - F1_reference), Leakage Sensitivity Index (LSI = β_overlap × 0.10), and Error-Reduction Illusion (ERI):

$$ERI = [\ \Delta F1 / (1 - F1_reference)\ ] \times 100\% \quad (2)$$

IV. RESULTS AND DISCUSSION

A. Effect of Controlled Duplicate Overlap (RQ1)

Controlled duplicate overlap produced massive, statistically significant F1 inflation in tree-based models, while leaving Logistic Regression largely unaffected (Table II).

TABLE II. F1 PERFORMANCE AND INFLATION AT 50% OVERLAP (0% MISSINGNESS)

Dataset	Model	Ref F1	50% Overlap	ΔF1	ERI (%)
Wine Red	**Decision Tree**	**0.697**	**0.873**	**+0.176***	**57.9%**
Wine Red	**Random Forest**	**0.777**	**0.913**	**+0.136***	**60.8%**
Wine Red	**Logistic Reg.**	**0.757**	**0.763**	**+0.006***	**2.6%**
Wine White	**Decision Tree**	**0.766**	**0.908**	**+0.142***	**60.5%**
Wine White	**Random Forest**	**0.834**	**0.932**	**+0.098***	**58.7%**
Wine White	**Logistic Reg.**	**0.820**	**0.820**	**+0.0004**	**0.3%**
Adult Income	**Decision Tree**	**0.615**	**0.817**	**+0.202***	**52.5%**
Adult Income	**Random Forest**	**0.679**	**0.847**	**+0.168***	**52.2%**
Adult Income	**Logistic Reg.**	**0.665**	**0.671**	**+0.005***	**1.6%**

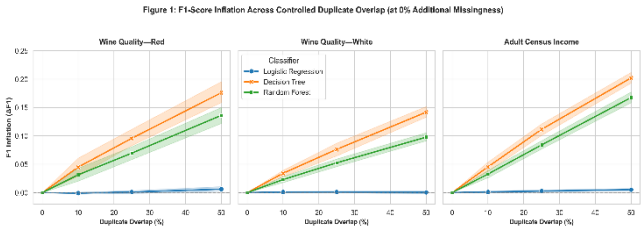


Fig. 1. F1-score inflation (ΔF1) across controlled duplicate overlap (0%, 10%, 25%, 50%) at 0% additional missingness with 95% confidence interval bands.

As shown in Fig. 1, tree models exhibit a linear dose-response sensitivity to duplicate overlap. Decision Tree F1 increased by +0.142 to +0.202, while Random Forest F1 rose by +0.098 to +0.168. In contrast, Logistic Regression remained essentially unaffected (ΔF1 ≤ +0.006). The Error-Reduction

Illusion reveals that 52% to 61% of residual classification errors appear to vanish purely from duplicate memorization.

This divergence stems from inductive bias: Decision trees recursively partition space into axis-aligned hyper-rectangles. A leaked test instance in training creates a pure leaf node matching the test label exactly, functioning as a nearest-neighbor hash lookup. Ensembles average these pure leaves across trees. Logistic regression optimizes a global hyper-plane; duplicating instances merely shifts sample weights slightly along the decision boundary without permitting localized memorization.

B. Interaction Between Missingness and Leakage (RQ2)

OLS regressions (Table III) demonstrate that the interaction between missingness and leakage is mediated by feature dimensionality.

TABLE III. OLS INTERACTION MODELS (ΔF1 ~ OVERLAP × MISSINGNESS)

Dataset	Model	β_overlap	β_missing	β_inter	p_inter
Wine Red	**Pooled**	**+0.203**	**-0.013**	**+0.044**	**0.659**
Wine White	**Pooled**	**+0.157**	**-0.009**	**-0.029**	**0.682**
Adult Income	**Decision Tree**	**+0.406**	**-0.007**	**+0.115**	**0.0137***
Adult Income	**Random Forest**	**+0.337**	**+0.009**	**+0.244**	**6.6e-10****

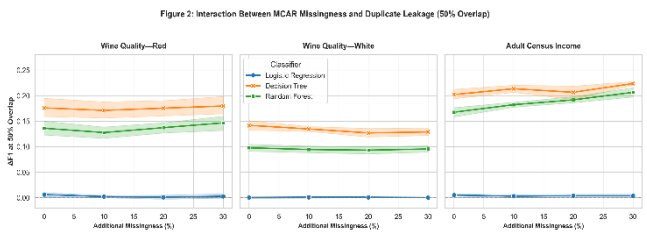


Fig. 2. Interaction between MCAR missingness and duplicate leakage at 50% overlap across models.

In the 11-feature Wine datasets, the interaction term is non-significant ($p > 0.60$), as median imputation preserves coordinate matching between paired copies. In Adult Census Income, missingness significantly amplifies duplicate leakage in tree models ($\beta_inter = +0.244$, $p < 10^{-9}$ for Random Forest; Fig. 2). As missingness degrades the reference baseline from 0.679 to 0.640, leaked duplicate test points act as imputed anchors that trees exploit, expanding net F1 inflation to +0.241 at 30% missingness.

C. Ordinary Random Splitting vs. Duplicate-Aware Splitting (RQ3)

Natural duplicate prevalence dictates the evaluation bias incurred by ordinary random splitting (Table IV and Fig. 3).

TABLE IV. EVALUATION BIAS IN ORDINARY RANDOM SPLITTING (0% MISSINGNESS)

Dataset	Natural Dups	Decision Tree ΔF1	Random Forest ΔF1	Logistic Reg. ΔF1
Wine Red	**15.0%**	**+0.058 ± 0.021****	**+0.035 ± 0.012***	**-0.008 ± 0.018**
Wine White	**19.1%**	**+0.066 ± 0.015****	**+0.047 ± 0.010****	**+0.005 ± 0.008**
Adult Income	**0.05%**	**-0.002 ± 0.012**	**-0.009 ± 0.014**	**-0.009 ± 0.014**

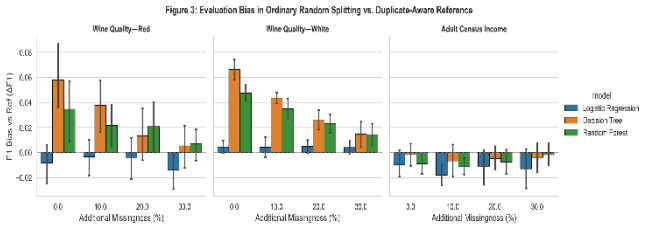


Fig. 3. Evaluation bias of ordinary random splitting compared to duplicate-aware reference across missingness levels.

In Adult, where natural duplicates constitute 0.05% of rows, ordinary random splitting introduces zero bias ($p = 0.922$). However, in Wine Red and White (15.0% and 19.1% duplicates), random splitting allows 21.4% and 27.6% of test records to have training twins, artificially inflating tree F1 by +0.035 to +0.066 ($p < 0.05$). Published studies using naive random splits on Wine Quality are systematically overestimating tree performance.

D. Exact Copies vs. Independently Masked Copies (RQ4)

The supplementary experiment decouples record presence from exact coordinate matching. At 0% missingness, independently masked copies retain 100% of exact-copy inflation. However, as missingness reaches 30%, retained leakage collapses to 19.5%–23.7% for Decision Trees and 15.0%–32.3% for Random Forests (Fig. 4).

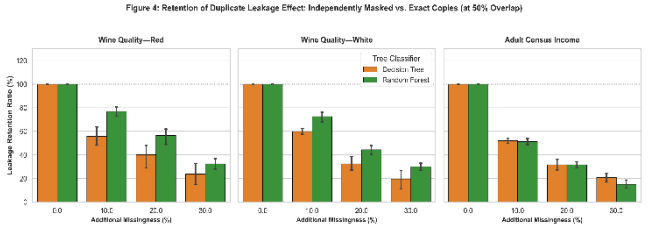


Fig. 4. Leakage retention ratio of independently masked copies vs. exact copies across missingness rates.

When masks are sampled independently, differing feature dropouts route the test twin down an alternative branch of the tree, destroying memorized leaf purity. This proves that

duplicate leakage is driven by exact multi-dimensional coordinate identity rather than local density.

E. Practical Recommendations for ML Engineering

We prescribe four concrete safeguards: (1) Mandatory Raw Duplicate Auditing before any transformation; (2) Enforce Group-Aware Splitting (StratifiedGroupKFold) so identical records never cross partitions; (3) Encapsulate Imputers within Pipeline objects fitted strictly on training folds; and (4) Audit Model Ranking Disparities, recognizing that sudden tree leads over linear baselines are frequent symptoms of duplicate contamination.

V. THREATS TO VALIDITY

Internal validity was protected by strict seed controls, non-parametric Wilcoxon testing, and leak-free pipeline encapsulation. External validity is bounded by our focus on three tabular benchmarks and three model architectures; future work should examine gradient-boosted trees and deep tabular architectures [7], [8]. Construct validity is bounded by our focus on MCAR missingness and exact duplicates; real-world environments may exhibit MAR/MNAR dynamics and near-duplicate entity fuzziness.

VI. CONCLUSION AND FUTURE WORK

This study systematically examined the interaction between missing values and duplicate leakage across 2,160 model evaluations. We demonstrated that duplicate leakage is highly model-dependent: flexible tree architectures suffer massive F1 inflation (up to +0.202 at 50% overlap), while linear models remain essentially immune. Standard random splitting on datasets with moderate natural duplicates (15%–19%) introduces statistically significant inflation (+0.035 to +0.066). In higher-dimensional data, missingness amplifies leakage by degrading the true signal while preserving duplicate memorization. Future work should investigate non-random missingness (MAR/MNAR) and soft record linkage in streaming environments.

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